Week 2
Describing Data

A summary is a decision about what to throw away.

Chapter 3 • Lecture

Mohammad Sabokrou
New Uzbekistan University

THE HABIT FOR TODAY

Never report a centre without a spread.

And never report either without knowing the shape they came from.

Course page

all slides, labs, book and data

Course page • sabokrou.github.io/teaching/data-science-fundamentals

SUMMARY CONTRACT

Four questions behind every summary

A number earns its place only when a reader can tell what it describes.

What is one row?	One student, one registration, or one student-week? The unit determines the denominator.
What is the unit?	Points, minutes, a count, or a category? Arithmetic needs a meaningful measurement scale.
How much data?	Report valid n and what is missing. A mean without a denominator can mislead.
What is the shape?	Check skew, groups, gaps, and unusual values. A plot tells you what a single number left out.

Mohammad Sabokrou
New Uzbekistan University

TEMPLATE

quantity + unit + population + valid n + centre + spread + shape or limitation

WHERE WE WERE

Last week: what a number is allowed to mean

A concern is not a question	until it names a population, a variable, a period and a decision
Evidence has a ceiling	a correlation supports a description, never an intervention
A row is a choice	4,665 registrations are not 4,665 students
The wrong unit biases in a direction	11.9% against 14.4%, with no warning either way

Mohammad Sabokrou
New Uzbekistan University

This week: what a number leaves out. Every summary discards information on purpose — the skill is knowing what you discarded.

CARRIED FORWARD

Every number today still needs a unit. “Mean what, of what, measured how?”

BEFORE CALCULATION

One row, one unit, one population

The word “average” means different things in different tables.

students.csv	1,200 rows: one student, entry-score points A mean describes students.
enrolments.csv	4,665 rows: one student-course registration, final-grade points A mean describes registrations unless you aggregate first.
activity.csv	16,165 rows: one student-week, minutes on platform A mean describes a student-week, not a student or a term.

Mohammad Sabokrou
New Uzbekistan University

CARRY FORWARD

Before using .mean(), say aloud: one row is one _____.

Missingness changes the denominator

A summary describes the values that were observed, not automatically the whole population.

Observed

midterm_satisfaction is present for 916 of 1,200 students.

Missing

284 students have no recorded satisfaction value. That is 23.7% of the cohort.

Defensible sentence

“Among 916 respondents, mean satisfaction was 3.5.” Do not silently claim this represents every student.

Mohammad Sabokrou
New Uzbekistan University

QUESTION

Which students are missing, and could that difference change the conclusion?

QUESTION

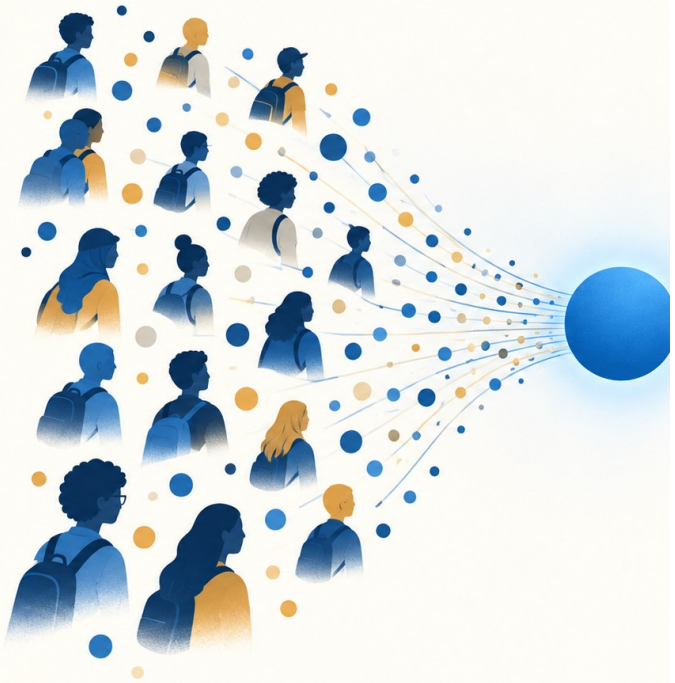
What disappears when 1,200 students become one average?

Look at the picture, then name two things a class average cannot tell you.

Think for 30 seconds

If the average entry score is 65.9, what do you still need to know before you make a decision?

Possible directions: spread, missing values, groups, unusual scores, or the unit being averaged.



Mohammad Sabokrou
New Uzbekistan University

PURPOSE

A summary is useful only after you know what it leaves out.

PART ONE

10–30

Statistics as compression

Mohammad Sabokrou
New Uzbekistan University

A summary replaces many numbers with few. That is useful and lossy in exactly the same act.

COMPRESSION

What compression erases

A summary is useful because it discards information. The risk is discarding the information the decision needs.

Individuals

Who is far from the centre? Who is missing? Who appears more than once?

Structure

Are there two groups, a long tail, a gap, or a change over time?

Context

What unit, period, collection process, and population does the value refer to?

Mohammad Sabokrou
New Uzbekistan University

DECISION TEST

If what you discarded could change the decision, the summary is not yet sufficient.

A SUMMARY IS A DECISION

1,200 numbers become one — what happened to the other 1,199?

What you gain

One number fits in a sentence, a headline, a table row.

It can be compared across groups, across years, across institutions.

It can be acted on.

What you lose

Every individual. The spread. The shape. The outliers. Whether it is one group or two.

All of it, silently, in one operation.

Mohammad Sabokrou
New Uzbekistan University

Compression is not a flaw in summarising. It is what summarising is. The question is always: is what I threw away relevant to the decision?

★ THINK

Name a decision where the spread matters more than the average.

CENTRE

Three answers to “what is typical?”

Mean — add them all, divide by how many

uses every value — and is moved by every value

Median — sort them, take the middle one

ignores how far away the extremes are

Mode — the most common value

the only one that works for categories

In our data

entry_score
mean 65.86
median 65.90

Almost identical — which itself tells you something about the shape.

When they diverge, that divergence is information, not a nuisance.

Mohammad Sabokrou
New Uzbekistan University

CHECK

programme has no mean and no median. Why, and what is left?

CENTRE

The mean is a balance point

The deviations from the mean add to zero. That is why every value can move it.

52, 58, 63, 71, 76 → mean = 64

Check the balance

Below 64: −12 − 6 − 1 = −19
Above 64: +7 +12 = +19

Mohammad Sabokrou
New Uzbekistan University

Why the mean moves

Change 76 to 760 and the mean becomes 200.8. One value changed, so the balance point moved with it.

INTERPRET

The mean uses every value. That is a strength when every value belongs, and a weakness when a value is wrong or unrepresentative.

CENTRE

The median is a position

Sort the values, then find the middle. Magnitude beyond the middle does not move it.

52, 58, 63, 71, 76 → median = 63

Extreme change

Replace 76 by 760.
Median remains 63.
Mean becomes 200.8.

Use the contrast

Mean and median disagree when a tail or extreme values pull the balance point away from the middle observation.

CAUTION

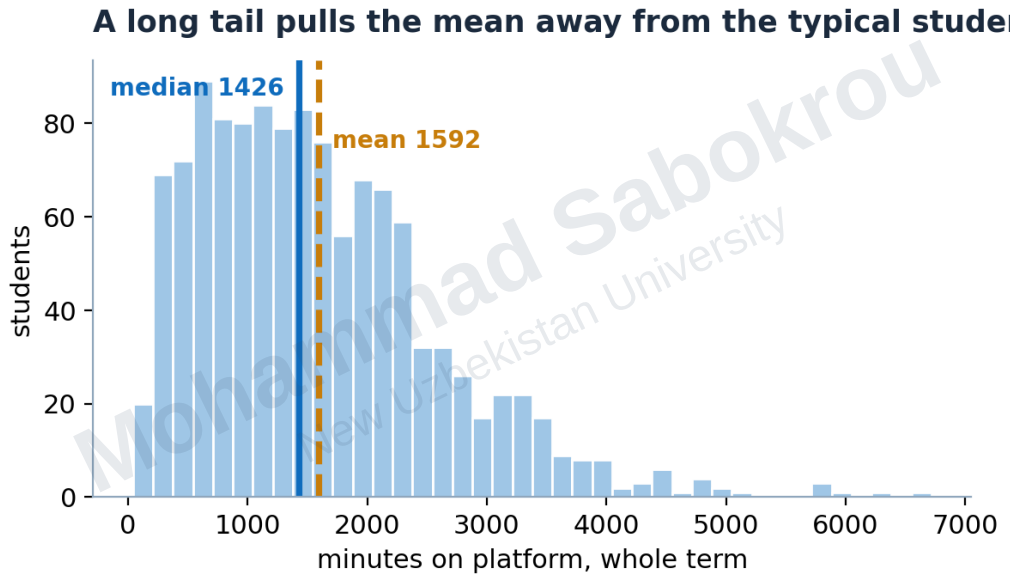
The median is robust to extremes, but it does not describe a distribution with two separate groups.

Mohammad Sabokrou
New Uzbekistan University

WHEN THEY DISAGREE

A long tail pulls the mean away from the typical student

Total minutes on the platform, per student, whole term. Same 1,200 students as last week.



Mohammad Sabokrou
New Uzbekistan University

Median 1,426. Mean 1,592. The gap is 166 minutes, and it is caused by a small number of very heavy users.

★ WHICH WOULD YOU REPORT?

The university wants to know what a typical student does. Mean or median — and what changes if you pick wrong?

CHOOSING CENTRE

Which centre answers this question?

The measure follows the data pattern and the decision. Each row uses a course-data example.

DATA PATTERN	STATISTICAL EXAMPLE	FIRST REPORT
Close to symmetric	entry_score: mean 65.86, median 65.9, skew −0.01	**Mean + SD are reasonable**
Long right tail	Platform minutes: mean 1,592, median 1,426, skew 1.10	**Median + IQR for a typical student**
Need a total	For total workload, mean × number of students estimates the sum	**Mean, even if the data are skewed**
A category	programme values are labels, not numerical distances	**Mode or a count table**
Suspect value	age = 305 needs investigation before a summary	**Check the record first**

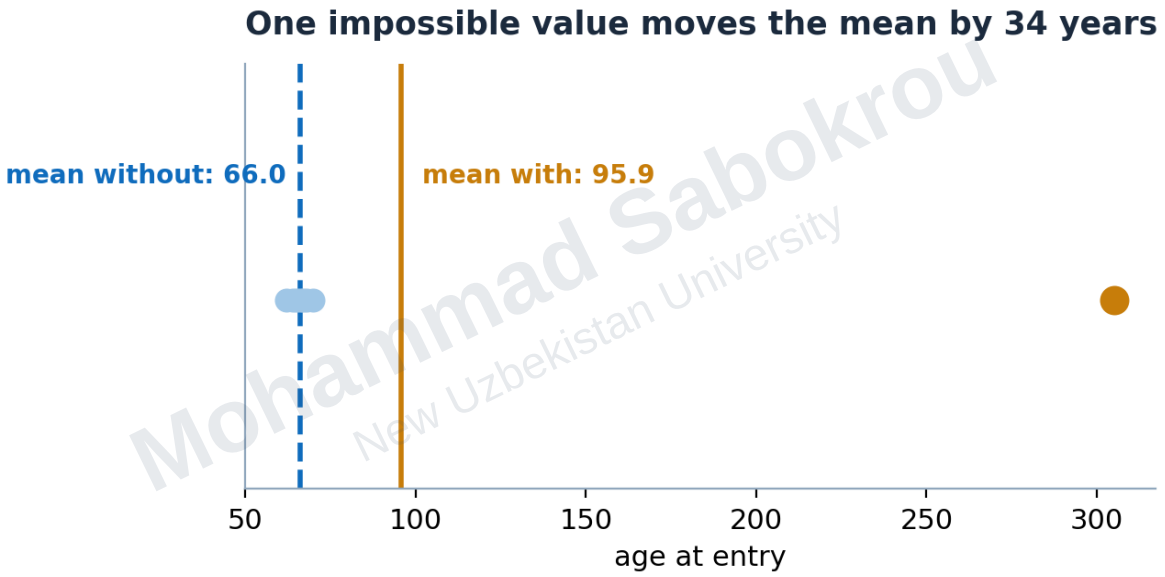
Mohammad Sabokrou
New Uzbekistan University

REPORT BOTH

When mean and median differ materially, report both and explain why they differ.

One impossible value moves the mean by 34 years

Seven real ages at entry, plus one value of 305 — which is really in students_dirty.csv, and which you will meet in Week 3.



Mohammad Sabokrou
New Uzbekistan University

The median of the eight values is 66.5, barely moved. The mean is wrecked.

★ THE LESSON IS NOT "USE THE MEDIAN"

It is: compute both, and if they disagree, go and look at the data before choosing.

Quantiles and robust summaries

20–34

When extreme values matter, the middle half of the distribution often describes the typical experience more honestly.

Mohammad Sabokrou
New Uzbekistan University

Percentiles start with an ordered list

A percentile tells you where a value sits after sorting. It does not automatically judge ability.

Mini-example: sort eight scores first

35	52	58	62	66	70	74	82	99
		Q1		Q2 / median		Q3		

Mohammad Sabokrou
New Uzbekistan University

Course data: 1,200 sorted entry scores

Q1 = 58.3 points

Median = 65.9 points

Q3 = 73.6 points

INTERPRET

At Q3 = 73.6, about 75% of observed entry scores are at or below 73.6.

QUANTILES

The five-number summary

A compact description that keeps the middle and the extremes visible.

Minimum	35.0 Lowest observed entry score
Q1, median, Q3	58.3 | 65.9 | 73.6 The middle half sits between Q1 and Q3.
Maximum	99.0 Highest observed entry score

Mohammad Sabokrou
New Uzbekistan University

USE

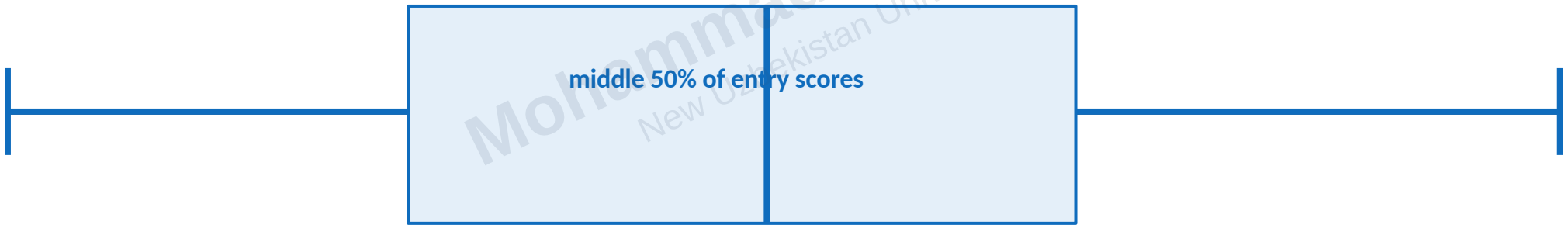
A five-number summary often pairs naturally with a boxplot and an explicit sample size.

The interquartile range keeps the middle 50% visible

IQR means interquartile range: Q3 − Q1. It is the spread used by the box in a boxplot.

Selected landmarks from the ordered entry-score column

min	Q1	median	Q3	max
35.0	58.3	65.9	73.6	99.0



Mohammad Sabokrou
New Uzbekistan University

IQR = Q3 − Q1 = 73.6 − 58.3 = 15.3 points

READ

A boxplot compresses the ordered column. It shows the middle half, the centre, and potential extremes at once.

OUTLIER CANDIDATES

IQR fences flag values for investigation

The calculation identifies unusual values. It does not decide what to delete.

1. Quartiles

Q1 = 58.3 Q3 = 73.6

2. Middle spread

IQR = Q3 − Q1 = 15.3

3. Fences

Lower = 35.35 Upper = 96.55

Mohammad Sabokrou
New Uzbekistan University

What the rule flags in this column

8 entry-score values fall outside the fences, including 35.0, 98.9, and 99.0.

Flagged means “inspect the source and context.” It does not mean “wrong” or “remove.”

RULE

Flag → inspect the record and measurement process → decide → document the reason.

OUTLIER HANDLING

Flag, investigate, document

Unusual values have several possible explanations. A deletion rule cannot distinguish them.

Wrong value	Age 305 in students_dirty.csv Check the original record and correct or remove only with evidence.
Valid extreme	6,717 total platform minutes Could be unusually high but valid engagement. Inspect, do not erase.
Different population	A value may be ordinary inside a subgroup Consider course, programme, period, and measurement process.

Mohammad Sabokrou
New Uzbekistan University

AUDIT TRAIL

Every correction or exclusion needs a reproducible rule and a written reason.

PART TWO

30–52

Spread

Mohammad Sabokrou
New Uzbekistan University

The centre tells you where. The spread tells you how much you should trust ‘where’ to describe anybody.

WHY CENTRE IS NOT ENOUGH

Three score sets with the same mean of 66

The average stays fixed, but the students' experience changes as the values spread out.

Class A	64 65 66 67 68	mean = 66 SD = 1.58
Class B	54 60 66 72 78	mean = 66 SD = 9.49
Class C	34 47 66 85 98	mean = 66 SD = 26.32

Mohammad Sabokrou
New Uzbekistan University

Same mean, very different spread. A centre without a spread does not describe the group.

SAY IT

A centre without a spread is not a summary. It is a rumour.

COMPARISONS

A comparison needs group size

A mean is not self-explanatory. Name the group and how many observations it contains.

CS101	n = 374 mean final grade = 65.5 SD = 10.7 A large course with moderate spread.
ST110	n = 229 mean final grade = 42.6 SD = 10.6 A different course with a different centre and similar within-course spread.
Whole cohort	n = 4,665 registrations mean final grade = 58.8 This aggregates courses with different grading distributions.

Mohammad Sabokrou
New Uzbekistan University

REPORTING HABIT

When comparing groups, state n, centre, spread, and the grouping rule together.

Four answers, increasing in usefulness

Range — max − min

uses only two values, and they are the two least typical ones

IQR — the middle 50%

ignores the extremes entirely — robust, and easy to explain

Variance — mean squared distance from the mean

in squared units, so not directly interpretable

Standard deviation — the square root of the variance

back in the original units — this is the one you report

entry_score

range 35 to 99

IQR 58.3 to 73.6
(a width of 15.3)

sd 10.94

The range spans 64 points. The middle half of students sit within 15.

Mohammad Sabokrou
New Uzbekistan University

NOTICE

The range grows as you collect more data. The IQR and the standard deviation do not.

A typical distance from the mean

entry_score has mean 65.9 and standard deviation 10.9. In plain language:

about two-thirds of students	sit between 55 and 77	*one sd either side*
about 95% of students	sit between 44 and 88	*two sd either side*
a student at 88	is two standard deviations above the mean	*unusual, but not extraordinary*

These percentages assume a roughly bell-shaped distribution. For the skewed minutes data they do not hold, and using them anyway is a real mistake.

★ TRY IT

final_grade has mean 58.8 and sd 13.7. Between which two values do about two-thirds of grades fall?

Mohammad Sabokrou
NUU Uzbekistan University

SPREAD

Standard deviation keeps the original unit

Variance is useful for calculation. Standard deviation is usually easier to communicate.

Variance

Average squared distance from the mean.

entry_score variance: 119.8 points²

The unit is squared.

Standard deviation

Square root of variance.

entry_score SD: 10.9 points

The original unit returns.

Plain language

A typical distance from the mean, when mean and SD are suitable for the distribution.

Mohammed Sabokrou
New Uzbekistan University

DO NOT SAY

Standard deviation is not a “typical error,” and it is not the standard error. We study standard errors in Week 8.

SPREAD

A range grows with more data

The range uses only two observations, so it becomes more likely to include an extreme as the sample grows.

Range	Good for showing observed limits Use it when the extreme itself matters, such as a safety threshold.
IQR	Good for describing the middle 50% Robust to a small number of extreme values.
Standard deviation	Good for distance around a meaningful mean Most informative when one distribution and a meaningful centre exist.

Mohammad Sabokrou
New Uzbekistan University

ALWAYS

Put n beside the range. Without n, readers cannot judge how much opportunity there was to observe an extreme.

WHY N − 1

The question every student asks, answered once

The short version

You measured the distances from the mean of your sample — not from the true mean of the population, which you do not know.

Your sample's own mean sits closer to your own data than the true mean would.

So what?

So the distances you measured are slightly too small, and dividing by n would give a spread slightly too low, every time.

Dividing by n − 1 corrects it.

Mohammad Sabokrou
New Uzbekistan University

With n = 1,200 the difference is invisible. With n = 5 it is not. The correction matters exactly when your data is small — which is when you are least likely to check.

IN PRACTICE

pandas uses n − 1 by default; numpy uses n. They will give you different answers on the same column.

IMPLEMENTATION

Software defaults must be named

Two valid formulas can produce different values when they answer different questions.

```
s = students['entry_score']

s.std()                 # pandas: n − 1
np.std(s)               # NumPy: n
np.std(s, ddof=1)       # matches pandas
```

Mohammad Sab
New Uzbekistan Univ

Why the difference

pandas defaults to the sample standard deviation. NumPy defaults to a population-style divisor.

What to write

State the library and ddof setting whenever another analyst could reproduce a different value.

LAB 2

Students compute both versions by hand and in code. The mismatch is the lesson, not a bug.

SHORT BREAK — TEN MINUTES

After the break: the shape that both numbers came from

Mohammad Sabokrou
New Uzbekistan University

Write down one variable from your own life that you think has a long tail.

SO FAR TODAY

mean 1,592
median 1,426

Same column. Both correct.
One of them describes people.

PART THREE

62–85

Shape

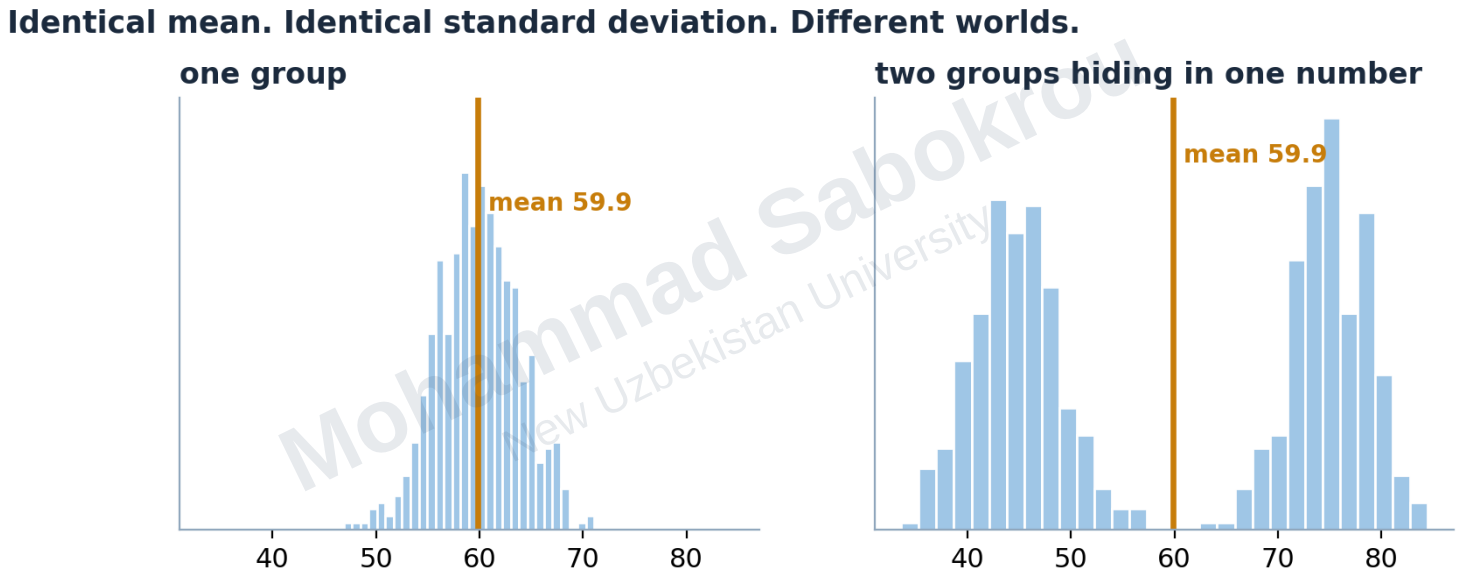
Mohammad Sabokrou
New Uzbekistan University

Two numbers cannot describe a distribution. Looking at it takes four seconds and prevents most of the mistakes in this chapter.

THE TRAP

Identical mean. Identical standard deviation. Different worlds.

Two synthetic groups, constructed to have the same mean and the same spread.



Mohammad Sabokrou
New Uzbekistan University

On the left, the mean describes almost everybody. On the right, it describes almost nobody — there is no student near 60.

★ EXPLAIN

If this were a course's grades, what would the right-hand picture tell you that the mean cannot?

SKEW

Which direction the tail runs

Left-skewed *tail runs low*

a few very small values pull the mean below the median

final_grade, skew −0.18

Roughly symmetric *no meaningful tail*

mean and median agree

entry_score, skew −0.01

Right-skewed *tail runs high*

a few very large values pull the mean above the median

minutes on platform, skew +1.10

Mohammad Sabokrou
New Uzbekistan University

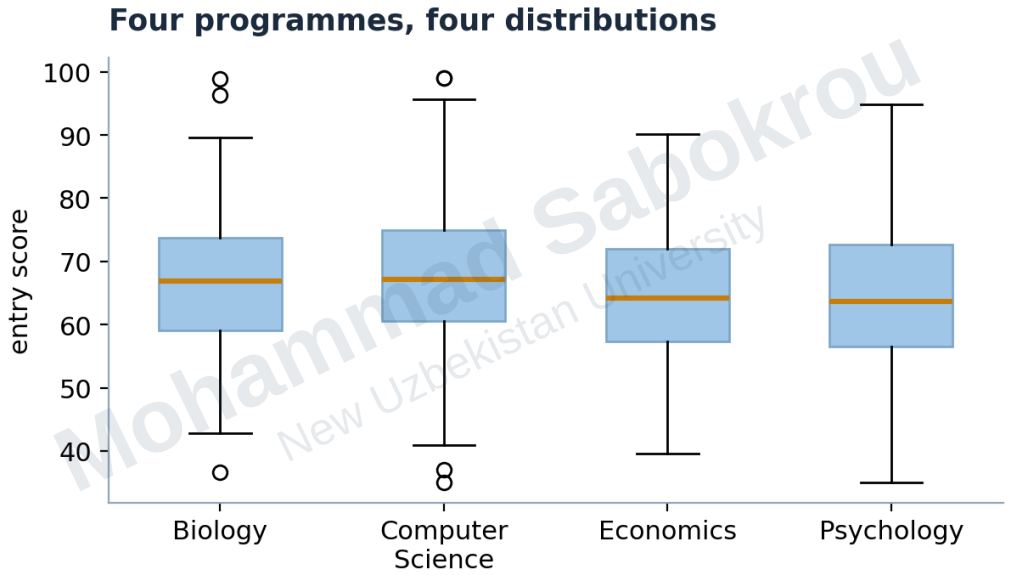
The quickest check needs no statistic at all: compare the mean with the median. If the mean is higher, the tail runs high.

PREDICT

Household income in any country. Which way does it skew, and which summary do governments usually report?

Four programmes, four distributions

A box plot shows the median, the middle 50%, and the outliers — four summaries at once, for four groups at once.



Mohammad Sabokrou
New Uzbekistan University

The medians are close. The spreads are not, and Psychology has the widest of the four.

★ READ IT

Which programme has the most consistent intake, and which has the widest range of preparation?

Histogram bins change the view

A histogram is a visual summary, and its bin width changes which patterns stand out.

Too few bins

Broad bars can hide a gap or two separate peaks.

Useful bins

A reasonable choice shows the main shape without turning every observation into noise.

Too many bins

Very narrow bars can make random variation look like a real pattern.

Mohammad Sabokrou
New Uzbekistan University

PRACTICE

Try more than one reasonable bin setting. If the conclusion changes, say so.

SHAPE

Inspect five features before choosing a summary

A quick visual check prevents a long list of later mistakes.

Centre	Where does most of the distribution sit? Mean, median, and mode answer different questions.
Spread	How far apart are typical values? Use IQR or SD in the original unit.
Skew and tails	Which direction stretches farther? Tails can pull the mean away from a typical case.
Modes and gaps	One population or hidden subgroups? A gap often asks for a grouping variable.
Unusual values	Data error, valid extreme, or a different process? Flag before you alter data.

Mohammad Sabokrou
New Uzbekistan University

ORDER

Plot first. Summarise second. Interpret third.

MULTIMODALITY

Two peaks ask a group question

When a distribution has distinct peaks, a single centre may lie where few observations exist.

Cohort mean = 58.8
DS100 mean = 63.1 MA110 mean = 45.4

What changed?

The courses have different grading distributions. The whole-cohort mean merges them into one number.

Mohammad Sabokrou
New Uzbekistan University

Better report

Name the grouping variable, report each group's n, centre, and spread, then explain what remains comparable.

DO NOT INFER

Different course means do not prove one course caused the difference. This is description, not causal evidence.

DISAGGREGATION

Disaggregate before comparing

A difference between mixed groups can reflect composition rather than a within-group difference.

Step 1

Start with the whole distribution.

This shows the practical question and the overall pattern.

Step 2

Choose a meaningful grouping variable before interpreting a gap.

Examples: course, programme, study mode, time period.

Step 3

Report group size and within-group spread.

A mean difference alone does not tell you whether the groups overlap.

Mohammad Sabokrou
New Uzbekistan University

LATER

Week 5 returns to this habit through confounding and Simpson’s paradox.

Normality is a model, not a setting

The normal distribution can be useful, but data do not become normal because software assumes it.

What a plot can do

Show an approximate shape, gaps, skew, and unusual values.

What a plot cannot do

Prove a distributional model from appearance alone.

What to do

State the assumption, inspect the data, and use a method suited to the question.

Mohammad Sabokrou
New Uzbekistan University

PREVIEW

Week 8 examines what normality means for intervals and tests. For now, do not apply normal rules automatically.

EMPIRICAL RULE

The 68–95 rule is conditional

It is a rough guide for roughly bell-shaped data, not a universal rule for every column.

about 68% within 1 SD
about 95% within 2 SD

Use carefully

entry_score is close to symmetric. Total platform minutes are right-skewed, so a median and IQR are more direct descriptions of a typical student.

Language

Say “about” and name the shape assumption. Do not turn a rule of thumb into a guarantee.

Mohammad Sabokrou
New Uzbekistan University

CHECK

If the histogram shows a long tail or two peaks, pause before quoting a normal-distribution percentage.

Outlier and impossible value are different claims

Both require investigation, but they call for different actions.

Unusual	A student with 6,717 platform minutes May be valid heavy use, a duplicated log, or a different usage pattern.
Impossible	Age 305 in a student registry Violates the data's meaning. Verify source, then correct or exclude with a documented rule.
In both cases	Keep the raw record and the decision trail A clean dataset without a rationale is not reproducible.

Mohammad Sabokrou
New Uzbekistan University

NEVER

Never use a generic z-score or IQR threshold as an automatic deletion button.

SHAPE PRACTICE

Diagnose before you summarise

Choose the summary that answers the decision, then state the limitation.

Which pair would you report first for a right-skewed income distribution? Which would you report first for a nearly symmetric machine measurement?

Use a centre and a spread. Explain why the distribution makes that choice sensible.

Mohammad Sabokrou
New Uzbekistan University

Right-skewed income

Median and IQR.
Explain the long high-value tail.

Symmetric measurement

Mean and SD.
Check that one group and a meaningful mean exist.

ANSWER QUALITY

A correct statistic without a reason is incomplete. The reason is the part that transfers to new data.

PART FOUR

85–105

Standardisation

Mohammad Sabokrou
New Uzbekistan University

How do you compare a score of 88 on one scale with 61 on another? You stop using the scale and start using the spread.

How many standard deviations from the mean?

```
z  =  (value − mean)  ÷  standard deviation
```

`z = 0`	exactly average
`z = +1`	one sd above — higher than about 84% of the group
`z = +2`	two sd above — unusual
`z = −1.5`	well below average

What it buys you

Two variables measured on completely different scales become comparable, because both are now expressed in units of their own spread.

A z of +2 means the same thing whether the original was a grade, a login count, or a temperature.

It is also the first step of almost every model you will build from Week 9.

★ COMPUTE ONE

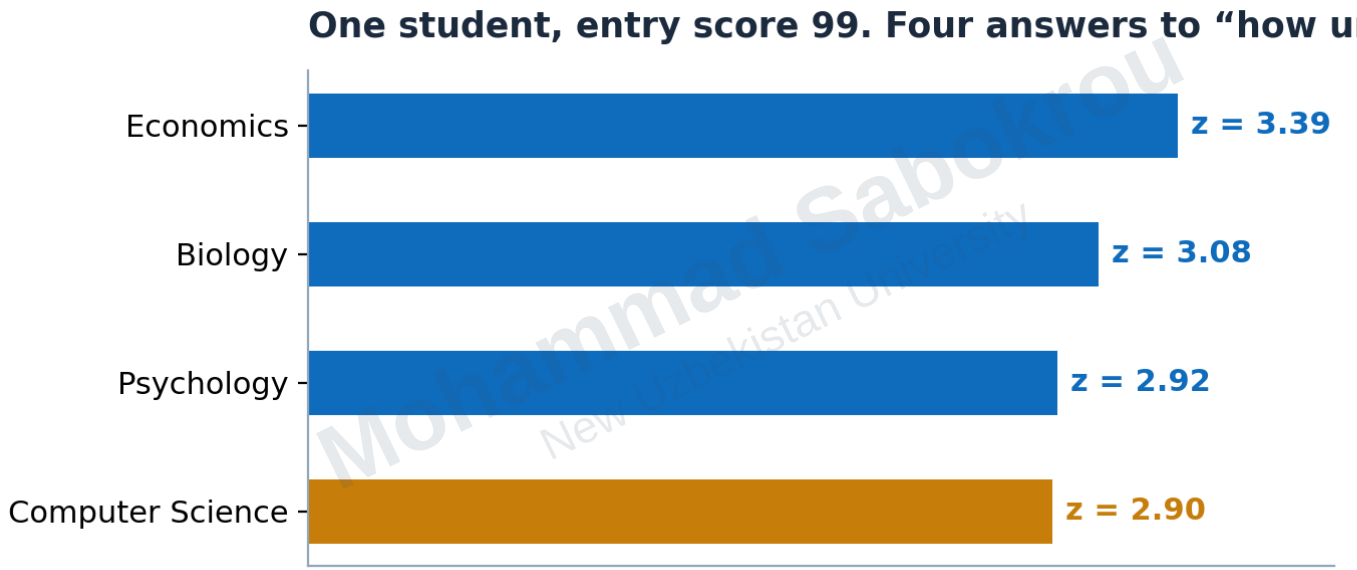
entry_score 88, mean 65.9, sd 10.9. What is z? Is that student unusual?

Mohammad Sabokrou
New Uzbekistan University

THE PART PEOPLE MISS

One student, entry score 99. Four answers to “how unusual?”

Student 100538 has the highest entry score in the dataset. How unusual that is depends entirely on who you compare them with.



Against Economics students, z = 3.39. Against their own programme, Computer Science, z = 2.90.

★ THE QUESTION TO ASK

A z-score is meaningless until you say: standardised against whom?

Mohammad Sabokrou
New Uzbekistan University

STANDARDISATION

Calculate the z-score from its reference group

A z-score is a distance from one chosen mean in units of that group's standard deviation.

$$z = (99 - 65.86) \div 10.94 = +3.03$$

Plain language

Student 100538's entry score is 3.03 standard deviations above the mean of all 1,200 students.

Say the reference

The sentence is incomplete without "standardised against the whole cohort."

DO NOT INFER

A large z-score describes position in a reference group. It does not explain why the score occurred.

Mohammad Sabokrou
New Uzbekistan University

REFERENCE GROUP

The same score changes with the comparator

Standardisation does not remove context. It makes the chosen context explicit.

Whole cohort	z = +3.03 Against all 1,200 students.
Computer Science	z = +2.90 Against the student's own programme.
Economics	z = +3.39 Against a different programme with a lower mean and different SD.

Mohammad Sabokrou
New Uzbekistan University

QUESTION

Whose comparison is relevant to the decision: cohort, programme, entry year, or something else?

RANK

A z-score is not a percentile

Distance and rank are different descriptions.

z = 1.5

The value sits 1.5 standard deviations above the chosen mean.

Percentile

The observed proportion at or below the value, or a model-based estimate under stated assumptions.

Good practice

Use the empirical rank when you have the distribution. State any model if you convert z to a percentile.

Mohammad Sabokrou
New Uzbekistan University

NEVER SKIP

Do not say “z = 1.5 means 93rd percentile” without naming the normal-distribution assumption or calculating the empirical rank.

MEASUREMENT

Standardise measurements, not arbitrary codes

A formula can produce a number even when the number has no meaning.

Entry score	0–100 points: z-score can be meaningful The distance in points has a defined interpretation.
Platform minutes	A meaningful unit, but inspect shape first A z-score may not give the clearest typical-case description for a long-tailed variable.
Programme code	1–4: do not standardise The arithmetic exists but the numerical distance between categories does not.

Mohammad Sabokrou
New Uzbekistan University

WEEK 1 RETURNS

Measurement level decides whether an operation is meaningful, not whether a library function accepts it.

Four questions, in order

01	**What decision does this number feed?**	Budget totals need the mean. “Typical” needs the median.
02	**What shape is the column?**	Plot it. One line of code, four seconds, prevents most errors on this slide.
03	**Could this be two groups?**	If the histogram has two lumps, no single summary is honest.
04	**What am I discarding, and does it matter?**	Name it in a sentence. That sentence is what separates a report from a number.

Course rule two, restated: a number without a sentence interpreting it is not an answer. Question 04 is that sentence.

FROM WEEK 3 ONWARD

Every summary you submit carries a sentence naming what it discarded. It is marked.

Mohammad Sabokrou
New Uzbekistan University

IN CODE

Everything from today, in eight lines

```
students["entry_score"].mean()        # 65.86
students["entry_score"].median()      # 65.90
students["entry_score"].std()         # 10.94
students["entry_score"].quantile(.25) # 58.3
students["entry_score"].skew()        # -0.01

students["entry_score"].describe()
students["entry_score"].hist(bins=30)

students.groupby("programme")["entry_score"].agg(
    ["count", "mean", "median", "std"])
```

Mohammad Sabokrou
New Uzbekistan University

Two warnings

describe() will happily summarise student_id. It is still meaningless.

numpy's std() divides by n; pandas divides by n − 1. On the same column they disagree. Use pandas, or pass ddof=1.

★ DO IT NOW

Run the groupby line. Which programme has the widest spread?

IN CODE

Profile one numeric column before modelling

A short, reproducible profile catches surprises before they become model features.

```
x = students['entry_score']

x.agg(['count', 'mean', 'median', 'std', 'min', 'max'])
x.quantile([.25, .50, .75])
x.hist(bins=20)
```

Mohammad Sab
New Uzbekistan Univ

Read the output

Count and quantiles answer different questions from mean and SD. Keep all of them in view.

Then interpret

Write one sentence about centre, one about spread, and one about shape or limitation.

REPRODUCIBILITY

The code is short. The explanation is the assessed work.

IN CODE

A group-wise profile finds hidden structure

Aggregate after deciding what one row should mean, then report the group definition.

```
(enrolments.groupby('course_code').final_grade
 .agg(n='count', mean='mean', median='median', sd='std')
 .sort_values('mean'))
```

Mohammad Sab
New Uzbekistan Univ

What this changes

The overall final-grade mean of 58.8 becomes a set of course-level distributions that can be inspected honestly.

What it cannot do

It cannot establish that a course caused the difference. It only makes the structure visible.

RULE

Do not choose subgroups only after seeing an interesting result. State why the group belongs in the analysis.

WRITING RESULTS

A defensible reporting sentence

The reader should not need to reconstruct the context from your code.

Among 1,200 students, entry scores had a mean of 65.9 points and a standard deviation of 10.9 points. The distribution was nearly symmetric; the middle half of scores lay from 58.3 to 73.6 points.

It states population, n, unit, centre, spread, shape, and an additional robust summary.

Mohammad Sabokrou
New Uzbekistan University

What it avoids

“Average score = 65.9.”
No unit, n, spread, or shape.

What remains

It still does not prove a cause, explain the scores, or describe missing data in another column.

WRITING RULE

A number without a sentence interpreting it is not an answer.

Five mistakes, all of them common

Reporting a mean with no spread	The single most common fault in student reports, and the easiest to fix.
Using the 68/95 rule on skewed data	It assumes a bell shape. Check the shape first.
Reporting the mean of an ordinal scale	“Mean satisfaction 3.4” treats unequal gaps as equal. Report the distribution too.
Summarising before plotting	Two lumps and one lump give the same mean. You cannot tell without looking.
Standardising without naming the reference group	z = 2.9 or z = 3.39 for the same student, depending on who you compared them with.

★ ONE HABIT PREVENTS FOUR OF THE FIVE

Plot the column before you summarise it.

Mohammad Sabokrou
New Uzbekistan University

What you can do now

A summary is compression	and you are responsible for what you discarded
Mean and median answer different questions	when they disagree, that disagreement is information
A centre without a spread is a rumour	10.9 is what makes 65.9 mean anything
Two numbers assume one lump	plot it, or you are guessing
A z-score needs a reference group	the same student is z = 2.90 or z = 3.39 depending on the comparison

Last week: what a number is allowed to mean. This week: what it leaves out. Next week: whether you can trust it at all.

THE HABIT

Never report a centre without a spread — and never report either without looking at the shape.

Mohammad Sabokrou
New Uzbekistan University

DECISION CASE

Choose a summary for platform use

The Dean asks: “What does a typical student’s platform use look like?”

Total platform minutes per student have mean 1,592, median 1,426, a long right tail, and 23 values above the 1.5 IQR upper fence.

Write a two-sentence answer for the Dean. Which centre and spread describe the typical student? What would you report separately for workload planning?

Mohammad Sabokrou
New Uzbekistan University

Typical student

Median 1,426 and IQR.
Explain the right-skewed distribution.

Capacity planning

Mean or upper quantiles may matter.
Explain why high-use students affect resource demand.

TRANSFER

The statistic follows the decision. There is no universally “best” centre.

EXIT TICKET

Four short answers before you leave

1. The mean of minutes on platform is 1,592 and the median is 1,426. Which would you report as ‘typical’, and why?

2. entry_score has mean 65.9 and sd 10.9. Between which values do about two-thirds of students sit?

3. Two courses both have a mean grade of 58. Name one thing that could still make them completely different.

4. A student has z = +2.1. State what that means, and name the one piece of information you need before you can interpret it.

Mohammad Sabokrou
New Uzbekistan University

★ HAND IN

On your way out. Bring your answers to Lab 2.

LAB 2 HANDOFF

Summaries by hand, then in code

By hand first

Eight numbers, on paper: mean, median, sd. Then check against pandas.

Deliverable

One notebook, plus a 150-word note on one column whose mean you would not report alone.

Mohammad Sabokrou
New Uzbekistan University

In the lab you will

- Compute mean, median and standard deviation by hand for a small set, then verify in pandas
- Plot four columns and describe the shape of each in one sentence
- Find one column where the mean and median disagree, and explain why
- Standardise one column and state explicitly which group it was standardised against

Carry this rule: never report a centre without a spread, and plot before you summarise.

Reading • Volume 1, Chapter 3